Science practice booklet: Grade 5 expectations

Grade 5 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 5 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: 5-LS1-1, 5-LS2-1

A worked example

In a mangrove forest, crabs eat fallen mangrove leaves and herons eat the crabs. Write the food chain, show which way the energy moves, and say where the energy first comes from.

1. Start with the producer: the mangrove tree makes its own food using sunlight.
2. The crab eats the leaves, and the heron eats the crab.
3. Write the chain with arrows pointing from the food to the one that eats it: mangrove leaves to crab to heron.
4. The energy first comes from the Sun, because the mangrove uses sunlight to make food.

Mangrove leaves to crab to heron; the energy first comes from the Sun.

Questions

1. Where does a plant get the energy it needs to make food?

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2. A date palm seedling grows into a tall tree, but the sand around its roots hardly loses any weight. Where does most of the tree's new material come from?

   A. The sand around its roots

   B. Water and carbon dioxide

   C. The light of the Sun itself

   D. Salt taken from the ground

3. What is a producer? Give one example from the UAE.

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4. What is a consumer? Is a camel a consumer? Explain.

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5. In a mangrove forest, crabs eat fallen leaves and herons eat the crabs. Which food chain is written correctly?

   A. Heron to crab to mangrove leaves

   B. Crab to mangrove leaves to heron

   C. Mangrove leaves to heron to crab

   D. Mangrove leaves to crab to heron

6. What do the arrows in a food chain show?

   A. The direction energy moves, from the food to the one that eats it

   B. Which animal is bigger

   C. The direction the animal walks

   D. The direction energy moves, from the eater to its food

7. In the desert, grass is eaten by a hare and the hare is eaten by a fox. Write this food chain with arrows.

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8. In the desert a fox eats hares and also lizards. What does this show about food webs?

   A. Foxes are producers

   B. Several food chains join into a food web because animals eat more than one food

   C. Hares are decomposers

   D. Energy moves from the fox to the grass

9. A fallen mangrove leaf rots in the mud and its nutrients go back to the soil. Which living things do this?

   A. Producers such as mangrove seedlings

   B. Consumers such as herons

   C. Decomposers such as fungi and bacteria

   D. Predators such as hawks

10. In a mangrove forest the number of crabs falls a lot. Name two things that might happen.

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11. Explain why almost all the energy in a food web can be traced back to the Sun.

Section 2: Earth and space

Standards practised: 5-ESS1-1, 5-ESS1-2, 5-ESS2-1, 5-ESS2-2, 5-ESS3-1

A worked example

Salem measures the shadow of a tall pole at three times of day. At 8 am it is 90 cm, at 12 noon it is 20 cm and at 4 pm it is 85 cm. When is the Sun highest in the sky, and why does the shadow change?

1. The shortest shadow is the one at 12 noon (20 cm).
2. A short shadow means the Sun is high in the sky.
3. The Earth spins, so the Sun seems to move across the sky during the day.
4. When the Sun is low (morning and afternoon) the shadow is long, and when it is high the shadow is short.

The Sun is highest at 12 noon, because the Earth spins and the Sun seems to move across the sky.

Questions

1. What causes day and night on Earth?

   A. The Sun moves around the Earth each day

   B. The Earth moves around the Sun each day

   C. The Earth spins on its axis once about every 24 hours

   D. The Moon covers the Sun at night

2. Hind measures the shadow of a pole at 7 am, at noon and at 4 pm. Which shadow is shortest, and why?

   A. Noon, because the Sun is highest in the sky

   B. 7 am, because the Sun is highest in the sky

   C. 4 pm, because the Sun is nearest to the Earth

   D. All the same, because the pole does not move

3. In the morning your shadow falls to the west. In which direction is the Sun?

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4. Why does the Sun look much bigger and brighter than the other stars?
   A. It is the biggest star in the universe
   B. It is hotter than every other star
   C. It is made of a different kind of matter
   D. It is much closer to the Earth

5. Why do we see different stars in the night sky in winter and in summer?

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6. In summer in the UAE the daylight lasts longer than in winter. What makes the length of daylight change through the year?

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7. Name the four Earth systems that interact with each other.

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8. Wind blows sand into a dune in the desert. Which two Earth systems are interacting?
   A. Biosphere and hydrosphere
   B. Atmosphere and geosphere
   C. Hydrosphere and biosphere
   D. Atmosphere and hydrosphere

9. Which statement about the water on Earth is true?
   A. Most of it is fresh water in rivers
   B. Half is salty and half is fresh
   C. Most of it is salty ocean water, and much of the fresh water is frozen
   D. Most of it is water vapour in the air

10. Why does the UAE make fresh water from sea water by desalination?

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11. Noura and her brother want to save water at home. Give two ways their family can do it.

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12. Explain how protecting mangrove forests helps both nature and people.

Section 3: Matter, forces and energy

Standards practised: 5-PS1-1, 5-PS1-2, 5-PS1-3, 5-PS1-4, 5-PS2-1, 5-PS3-1

A worked example

Maryam stirs 30 g of salt into 200 g of water until the salt seems to disappear. What is the mass of the salty water? How could she get the salt back?

1. The salt is still there. It has broken into particles too small to see and spread through the water.
2. Matter is not lost when it dissolves, so the masses add up: 200 g + 30 g = 230 g.
3. She can heat the salty water gently so the water evaporates; the salt stays in the pot.

230 g; she can evaporate the water and the salt is left.

Questions

1. Air is invisible. Which fact shows that air is matter?

   A. It takes up space and has mass

   B. It has a colour and a smell

   C. It never changes when it is heated

   D. It cannot be squashed or moved

2. Matter is made of particles that are too small to see. Use this idea to explain how you can smell karak from across a room.

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3. Hamdan dissolves 50 g of sugar in 250 g of water. What is the total mass of the sugar water?

   A. 250 g

   B. 200 g

   C. 50 g

   D. 300 g

4. An ice cube has a mass of 40 g. It melts completely. What is the mass of the water?

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5. Why is the handle of a cooking pot made of wood or plastic and not metal?

   A. It conducts heat well, so the food cooks faster

   B. It conducts heat poorly, so it stays cool enough to hold

   C. It is magnetic, so it holds the pot

   D. It dissolves in water, so it is easy to wash

6. Salt and sand are mixed together. Describe one way to separate them.

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7. Desalination plants make fresh water from sea water. Is sea water a mixture? How can the water be separated from the salt?

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8. Which of these is a chemical change?

   A. Ice melting in a glass

   B. Sugar dissolving in karak

   C. A sheet of paper torn into pieces

   D. An iron nail rusting

9. Baking soda is added to vinegar and bubbles of gas form. Is this a physical change or a chemical change? How can you tell?

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10. A child in the UAE and a child on the opposite side of the world each drop a stone. Which way does gravity pull both stones?

    A. Towards the Sun

    B. Straight up for the child far away

    C. Towards the centre of the Earth

    D. Towards the nearest ocean

11. The energy in the dates Layla eats first came from the Sun. Explain how.

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12. A garden light at Omar's grandfather's house is charged by a small solar panel during the day and glows at night. Where did the energy for the light first come from?

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13. Noura stirs sugar into her karak and says the sugar has disappeared, so there is less matter in the cup. Explain what is really happening.

Answer guide for parents

Section 1: Living things

1. The Sun gives the light energy a plant uses to make food, so the answer is the Sun.
2. B. Water and carbon dioxide. The tree joins water and carbon dioxide into new material using sunlight, so the sand barely changes. If your child chose A, they think plants eat soil.
3. A producer makes its own food using sunlight, so a mangrove tree, a ghaf tree or a date palm all count.
4. A consumer cannot make food, so it eats other living things. A camel eats plants, so it is a consumer.
5. D. Mangrove leaves to crab to heron. Arrows point from the food to the one that eats it, so leaves go to crab and crab goes to heron. If your child chose A, they pointed the arrows from the eater to its food.
6. A. The direction energy moves, from the food to the one that eats it. Arrows show energy moving from the food to the animal that eats it. If your child chose D, they pointed the energy the wrong way round.
7. The hare eats the grass and the fox eats the hare, so the chain is grass to hare to fox. The arrows point from food to eater.
8. B. Several food chains join into a food web because animals eat more than one food. The fox eats two foods, so its chain crosses another chain and together they make a web. If your child chose A, they forgot that producers make their own food and foxes do not.
9. C. Decomposers such as fungi and bacteria. Fungi and bacteria break down dead things and return nutrients to the soil. If your child chose B, they mixed up eating living things with breaking down dead ones.
10. Herons eat crabs, so they lose food. Fewer crabs also eat fewer leaves, so leaves may pile up. (Any sensible answer.)
11. Plants use sunlight to make food. Animals eat plants, or animals that ate plants, so the energy started with the Sun.

Section 2: Earth and space

1. C. The Earth spins on its axis once about every 24 hours. The Earth spins once in about 24 hours, so each side faces the Sun and then faces away. If your child chose A, they think the Sun moves around us, which is how it looks.
2. A. Noon, because the Sun is highest in the sky. The Sun is highest at noon, so the shadow is shortest then. If your child chose B, they mixed up the low morning Sun with the high midday Sun.
3. A shadow falls on the side opposite the Sun. The shadow points west, so the Sun is in the east.
4. D. It is much closer to the Earth. All stars shine, but the Sun is the closest one, so it looks biggest and brightest. If your child chose A, they think it is bigger in space, but it only looks bigger because it is near.
5. The Earth travels around the Sun during the year. At night we face a different part of the sky each season, so we see different stars.

6. The Earth is tilted. In summer the UAE leans towards the Sun, so the Sun is up for more hours and the days are longer.
7. Geosphere is rock and land, hydrosphere is water, atmosphere is air and biosphere is living things. Those are the four.
8. B. Atmosphere and geosphere. Wind is air, which is the atmosphere, and sand is land, which is the geosphere. If your child chose D, they got the wind right but forgot that sand is land, not water.
9. C. Most of it is salty ocean water, and much of the fresh water is frozen. About 97 of every 100 parts of Earth's water are salty ocean water, and much fresh water is frozen. If your child chose A, they think rivers hold most of the water, but they hold very little.
10. The UAE has little rain and no big rivers, but sea water is plentiful. So the sea is turned into fresh water.
11. Use less water. For example, turn off the tap while brushing teeth and fix leaking taps. (Any two sensible ways.)
12. Mangroves shelter fish, crabs and birds, so nature benefits. Their roots slow the waves and stop the shore wearing away, so people benefit too.

Section 3: Matter, forces and energy

1. A. It takes up space and has mass. Matter takes up space and has mass, and a balloon full of air shows both. If your child chose B, they think matter must have a colour or smell, but air has neither.
2. Tiny scent particles leave the cup and spread through the air, which is a gas, to your nose. We cannot see them, but we can smell them.
3. D. 300 g. Nothing is lost when sugar dissolves, so 50 g + 250 g = 300 g. If your child chose A, they thought the sugar vanished and only the water counts.
4. Melting changes the state, not the amount of matter, so the water is still 40 g.
5. B. It conducts heat poorly, so it stays cool enough to hold. Wood and plastic pass heat on slowly, so the handle stays cool enough to hold. If your child chose A, they think a handle should carry heat well, but that would burn the hand.
6. Water dissolves the salt but not the sand. Filter out the sand, then boil away the water to get the salt back.
7. Yes, sea water is salt mixed with water. Heating turns the water to vapour, cooling turns it back to fresh water, and the salt is left in the pot.
8. D. An iron nail rusting. Rusting makes a new substance, rust, so it is a chemical change. If your child chose B, they think dissolving makes something new, but the sugar is still there.
9. Bubbles of gas show that a new substance was made. That makes it a chemical change.
10. C. Towards the centre of the Earth. Gravity always pulls towards the centre of the Earth, so both stones fall down. If your child chose B, they forgot that down means towards the Earth's centre for everyone.
11. The date palm used sunlight to make food. Layla eats the dates, so her body uses that energy from the Sun.
12. The panel collects energy from sunlight in the day. The light gives out that stored energy at night, so it began with the Sun.
13. The sugar breaks into tiny particles that spread through the karak. No matter is lost, so the mass of the cup stays the same.